

# Nikhil Kumar Singh

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## CONTACT INFORMATION

Room 515, Building G 26  
Campus Kaiserslautern

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*GitHub:* <https://github.com/ciniks117>

## CURRENT POSITION

Postdoctoral Researcher  
Max Planck Institute for Software Systems, Germany  
September 2025 – Present

## EDUCATION

**Indian Institute of Technology, Kanpur,**  
PhD, Computer Science and Engineering, Jan 2020 - Sept 2024 (9.50/10)  
Thesis : Synthesis of Safe and Efficient Feedback Controllers for Complex Dynamical Systems.  
Advisor: Dr. Indranil Saha

**Indian Institute of Technology, Kanpur,**  
MS, Computer Science and Engineering, Jan 2018 - Dec 2019 (9.0/10)  
Thesis : Specification guided Automated Debugging of CPS Models.  
Advisor: Dr. Indranil Saha

**National Institute of Technology, Jamshedpur,**  
B.Tech, Computer Science and Engineering, Aug 2010 - May 2014 (8.70/10)  
Project: Intelligent Web Search Engine.

## RESEARCH INTERESTS

Reinforcement Learning  
Robotics and Automation  
Formal Methods and Software Engineering  
Control Theory

## EXPERIENCE

**FARE Fellow,**  
IIT Kanpur  
Safe Reinforcement Learning based Controller Synthesis. **Jan'25 - Aug'25**

**Visiting Researcher,**  
Max Planck Institute for Software Ssystems, Germany  
Safe Learning using Kernel based Embeddings. **Nov'24 - Jan'24**

**Senior Student Research Associate**  
FMSAFE Project, IIT Kanpur  
*Sponsor: MHRD IMPRINT* **Jan'18 - Dec'19**  
Developed an Automated Tool for Debugging CPS Models.

**Software Engineer**  
OptumSoft, Bangalore **June'14 - Jun'16**  
Software Development for Distributed Systems.

## PUBLICATIONS

**Nikhil Kumar Singh,** Indranil Saha. “*Frugal Actor-Critic: Sample Efficient Off-Policy Deep Reinforcement Learning Using Unique Experiences.*” International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2024), Auckland, New Zealand, May 06-10, 2024. <https://dl.acm.org/doi/10.5555/3635637.3663037>.  
Code at <https://github.com/iitkcpslab/FAC>

**Nikhil Kumar Singh,** Indranil Saha. “*STL-Based Synthesis of Feedback Controllers Using Reinforcement Learning.*” AAAI Conference on Artificial Intelligence (AAAI 2023), Washington

D.C., February 07-14, 2023. <https://ojs.aaai.org/index.php/AAAI/article/view/26764>.  
Code at <https://github.com/iitkcpslab/rlstl>

**Nikhil Kumar Singh**, Indranil Saha. “*Specification Guided Automated Synthesis of Feedback Controllers.*” ACM SIGBED International Conference on Embedded Software (EMSOFT 2021), ACM/IEEE, virtual conference, October 08-15, 2021.

Published in ACM Transactions on Embedded Computing Systems (2021) <https://dl.acm.org/doi/10.1145/3477011>.

Code at <https://github.com/iitkcpslab/SCoSyn>

**Nikhil Kumar Singh**, Indranil Saha. “*Specification Guided Automated Debugging of CPS Models.*” International Conference on Embedded Software (EMSOFT2020), ACM/IEEE, virtual conference, September 20-25, 2020.

Published in IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (2020) <https://ieeexplore.ieee.org/document/9211574/>.

Code at <https://github.com/iitkcpslab/Blars>

\* The author names appear in lexicographic order.

## AWARDS & RECOGNITIONS

**Fellowship for Academic and Research Excellence** (Jan’25 - Aug’25).

**Prime Minister’s Research Fellowship** (Dec’20 - Dec-23).

**ACM India IARCS** travel grant (2024).

Invited to **Google Research Week 2023 & 2024**.

Nominated by the Department of CSE, IIT Kanpur for **Cadence Gold Medal** and **SIIC Student Innovation Award** for MS thesis.

**First** prize in Data structure & algorithm based event *CODATHON* in OJASS’13.

**Third** prize in Coding based event *CODEMANIA* in Ojass’ 13.

**Second** prize in the event *CODESTORM* conducted by the Computer Society of India at NIT-Jamshedpur held in March-2012.

**First** prize in the event *LET US C* conducted by the Computer Society of India at NIT-Jamshedpur held in April-2011.

## TALKS

|                                                                                                              |                                         |               |
|--------------------------------------------------------------------------------------------------------------|-----------------------------------------|---------------|
| <b>Frugal Actor-Critic: Sample Efficient Off-Policy Deep Reinforcement Learning Using Unique Experiences</b> | <i>AAMAS-24, Auckland, New Zealand,</i> | <b>May’24</b> |
| <b>Specification Guided Controller Synthesis Using RL</b>                                                    | <i>CSE Dept. IIT Kanpur</i>             | <b>Oct’23</b> |
| <b>Learning-based Controller Synthesis</b>                                                                   | <i>Ericsson AI Research</i>             | <b>Mar’23</b> |
| <b>STL-Based Synthesis of Feedback Controllers Using Reinforcement Learning</b>                              | <i>AAAI-23, Washington D.C.</i>         | <b>Feb’23</b> |
| <b>Introduction to Controller Synthesis</b>                                                                  | <i>CSE Dept. IIT Kanpur</i>             | <b>Oct’21</b> |
| <b>Specification Guided Automated Synthesis of Feedback Controllers</b>                                      | <i>EMSOFT-21, Online</i>                | <b>Oct’21</b> |
| <b>Specification-Guided Automated Debugging of CPS Models</b>                                                | <i>EMSOFT-20, Online</i>                | <b>Sep’20</b> |

## TEACHING

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|---------------------------------------------------------------------------|---------------------------------------------------------|------------------------|
| <b>Instructor</b> Introduction to PyTorch (PMRF Lecture series)           | <i>Institute of Smart Structures and Systems (ISSS)</i> | <b>Sep’23 - Dec’23</b> |
| <b>Instructor</b> Controller Synthesis using Python (PMRF Lecture series) | <i>Institute of Smart Structures and Systems (ISSS)</i> | <b>Mar’23 - Jul’23</b> |

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| <b>Instructor</b> Introduction to Controller Synthesis (PMRF Lecture series)<br><i>Institute of Smart Structures and Systems (ISSS)</i> | <b>Sep'22 - Dec'22</b>  |
| <b>Tutor</b> Introduction to Computing (ESC101)<br><i>Indian Institute of Technology, Kanpur</i>                                        | <b>May'22 - Jun'22</b>  |
| <b>Teaching Assistant</b> Software Development (CS253)<br><i>Indian Institute of Technology, Kanpur</i>                                 | <b>Jan'22 - May'22</b>  |
| <b>Teaching Assistant</b> Autonomous Cyber-Physical Systems (CS659A)<br><i>Indian Institute of Technology, Kanpur</i>                   | <b>Jan'21 - May'21</b>  |
| <b>Teaching Assistant</b> Embedded and Cyber-Physical Systems (CS637A)<br><i>Indian Institute of Technology, Kanpur</i>                 | <b>Sept'20 - Dec'20</b> |
| <b>Teaching Assistant</b> Formal Methods for Robotics and Automation (CS638A)<br><i>Indian Institute of Technology, Kanpur</i>          | <b>Jan'20 - May'20</b>  |

PROFESSIONAL  
SERVICES

**PC Member** ECAI-2025, AAAI-2025  
**Reviewer** ICC-2024